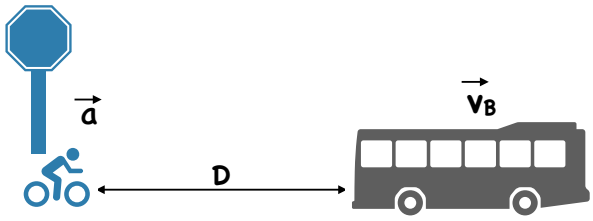
Problem 1 (20 points)

You are waiting at a bus stop with your bicycle. A bus moving at a constant speed of $v_B = 6$ m/s passes you. When the rear of the bus is $D = 27$ m past you, you start to ride toward it with a constant acceleration of $a = 2\ \mathrm{m/s^2}$.



(a) (10 pts.) Find the distance you would have to ride before you catch up with the rear of the bus.

If we set the origin (x=0) as the bus stop (your initial position), positions can be written as functions of time as follows:

$$x_{\text{you}}(t) = \frac{1}{2}at^2$$

$$x_{\text{bus}}(t) = 27 + v_{\text{bus}}\ t$$

$$x_{\text{bus}}(t') = x_{\text{you}}(t') \Rightarrow \frac{1}{2}(2)t'^2 = 27 + (6)\ t$$

$$t'^2 - 6t' - 27 = 0$$

$$t'_1 = -3\ \text{s},\ t'_2 = 9\ \text{s}$$

$$x_{\text{you}}(t) = \frac{1}{2}(2)(9)^2 = 81\ \text{m}$$

$d=$

(b) (10 pts.) Find your speed at the instant when you catch up with the rear of the bus.

$$v_{\text{you}}(t) = at$$

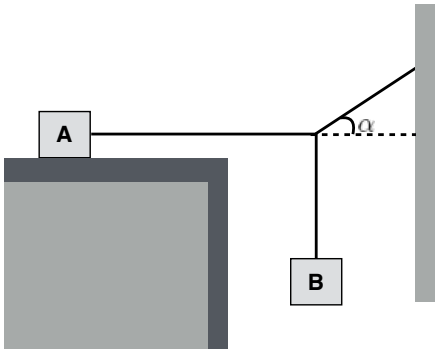
$$v_{\text{you}}(9\ \text{s}) = (2)(9) = 18\ \text{m/s}$$

$v=$

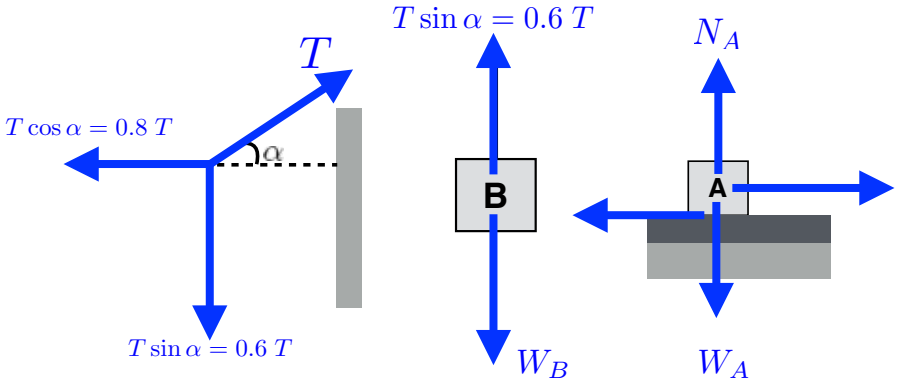
Problem 2 (20 points)

The system shown in the figure is in equilibrium. The weights of the blocks are given as $w_A = 80$ N and $w_B = 30$ N. The coefficient of static friction between block A and the surface it rests on is $\mu_s = 0.60$ and the angle shown is $\alpha = 36.9°$.

(You may take g=10 m/s^2.)



(a) (10 pts.) Find the friction force exerted on block A.



$$0.6\,T = W_B$$

$$T = \frac{10}{6}\,W_B$$

$$T\cos\alpha = \frac{4}{3}\,W_B$$

$$f_s = T\sin\alpha = \frac{4}{3}\,(W_B) = \frac{4}{3}\,(30\ N) = 40\ N$$

$F=$

(b) (10 pts.) Find the maximum weight of block B for which the system will stay in equilibrium.

$$f_s = \frac{4}{3}\,(W_B) \Rightarrow W_B = \frac{3}{4} f_s$$

$$\frac{3}{4} f_s^{\text{max}} = W_B^{\text{max}}$$

$$\frac{3}{4} f_s^{\text{max}} = \frac{3}{4}\mu_s N_A = W_B^{\text{max}}$$

$$W_B^{\text{max}} = \frac{3}{4}(0.6)(80) = 36\ N$$

$w_{B/max}=$

Problem 3 (20 points)

A disk of radius $r = 0.1$ m is rotating at $t = 0$ with an initial angular speed of 4π rad/s. It turns through 8 revolutions as it comes to stop with a constant angular acceleration.

(a) (5 pts.) Find the angular acceleration of the disk. Express your answer in terms of π and in units of rad/s^2.

$$\omega_f^2 - \omega_i^2 = 2\alpha\ \Delta\theta$$
$$0^2 - (4\pi)^2 = 2\ \alpha\ (8.2\pi)$$
$$\alpha = -\frac{\pi}{2}$$

$\alpha=$

(b) (5 pts.) Find the time for the disk to come to rest.

$$\omega_f - \omega_i = \alpha\Delta t$$
$$0 - 4\pi = (-\frac{\pi}{2})\Delta t$$
$$\Delta t = 8\ s$$

$t=$

(c) (5 pts.) Find the tangential acceleration of a point on the rim of the disk at $t = 2$ s. Express your answer in terms of π.

$$a_T = r\alpha = 0.1(-\pi/2) = -0.05\pi\ \text{m/s}^2$$

$a_{\text{tan}}=$

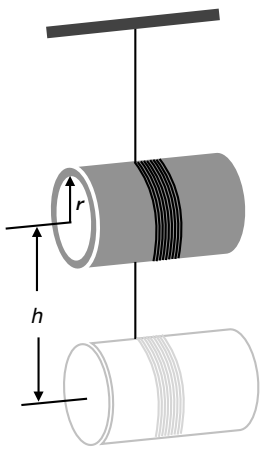
(c) (5 pts.) Find the radial acceleration of a point on the rim of the disk at $t = 2$ s. Express your answer in terms of π.

$$a_R(t) = r\omega^2(t) = r(\omega_i + \alpha t)^2$$
$$a_R(2) = r\omega^2(2) = (0.1)(4\pi - (\pi/2)(2))^2 = 0.9\ \pi^2\ \text{m/s}^2$$

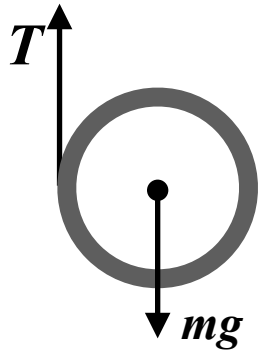
$a_{\text{rad}}=$

Problem 4 (30 points)

A string is wound around a uniform, thin-walled hollow cylinder of radius r and mass m. The cylinder is released from rest with string vertical and its top end tied to a fixed bar. ($I_{\text{cylinder}} = m\,r^2$)



(a) (10 pts.) Find the tension in the string in terms of the weight of the cylinder, m.



$$mg - T = ma$$
$$a = r\alpha$$
$$Tr = I\alpha = mr^2\alpha$$

$$Tr = mr(r\alpha) = mra$$
$$T = ma$$
$$mg - T = T \Rightarrow T = \frac{mg}{2}$$

$T=$

(b) (10 pts.) Find the magnitude of the acceleration of the center of mass in terms of gravitational acceleration g.

$$mg - T = ma$$
$$mg - \frac{mg}{2} = ma$$
$$a = \frac{g}{2}$$

$a=$

(c) (10 pts.) Find the speed of the center of mass after the cylinder has descended through a distance h (in terms of g and h).

Solution 1:

$$v_f^2 - v_i^2 = 2a\Delta y$$
$$v_f^2 - 0^2 = 2(\frac{g}{2})h$$
$$v_f^2 = \sqrt{gh}$$

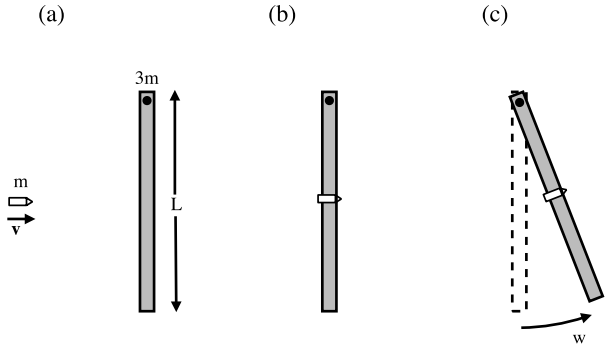
Solution 2:

$$mgh = \frac{1}{2}mv_f^2 + \frac{1}{2}I\omega_f^2$$
$$v_f = r\omega_f$$
$$mgh = \frac{1}{2}mv_f^2 + \frac{1}{2}mr^2\omega_f^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}mv_f^2$$
$$mgh = mv_f^2$$
$$v_f = \sqrt{gh}$$

$v_{\text{cm}}=$

Problem 5 (20 points)

A uniform rod of length L rests on a frictionless horizontal surface. The rod pivots about a fixed frictionless axis at one end. The rod is initially at rest. A bullet traveling parallel to the horizontal surface and perpendicular to the rod with speed v strikes the rod at its center and becomes embedded in it. The mass of the bullet is one-third the mass of the rod. Treat the bullet as a point object. ($I_{\text{rod}} = 1/3\, M\, R^2$)



(a) (5 pts.) Find the magnitude of the bullet's angular momentum with respect to the pivot point just before it strikes the rod.

$$L^i_{\text{bullet}} = mv\frac{L}{2}$$

$L=$

(b) (10 pts.) Find the angular speed of the rod just after the bullet becomes embedded in it.

$$L^i_{Bullet} = L^f_{Total} = I_{Total}\,\omega = \left(\frac{1}{3}(3m)L^2 + m(\frac{L}{2})^2\right)\omega$$

$$mv\frac{L}{2} = \frac{5}{4}mL^2\omega$$

$$\omega = \frac{2v}{5L}$$

$w=$

(c) (5 pts.) Find the ratio of the kinetic energy of the system just after the collision to the kinetic energy of the bullet before the collision.

$$K_i = \frac{1}{2}mv^2$$

$$K_f = \frac{1}{2}I_{Total}\,\omega^2 = \frac{1}{2}(\frac{5}{4}mL^2)(\frac{2v}{5L})^2 = \frac{1}{10}mv^2$$

$$r = \frac{K_f}{K_i} = \frac{(1/10)mv^2}{(1/2)mv^2} = \frac{1}{5}$$